

MEDICARE HOSPITAL VALUE ANALYSIS

What makes a high-value hospital?

CMS star ratings don't tell the whole story.

SQL, Python, Tableau, Databricks · 7 public CMS datasets · ~5,000 U.S. hospitals

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github.com/KabiraL/Hospital_Assessment



Which hospitals deliver higher outcomes at lower cost?

What sets them apart?

You don't get what you pay for

Outcomes and cost barely move together, so the best-value hospitals are efficient **exceptions** and not the rule.

Cost predicts; quality doesn't

A model flags likely low-cost hospitals with ~84% precision, yet clinical outcomes resist prediction from this data.

Stars miss hidden value

CMS star ratings ignore cost, so **2 in 5** high-value hospitals hide at just 1–3 stars.

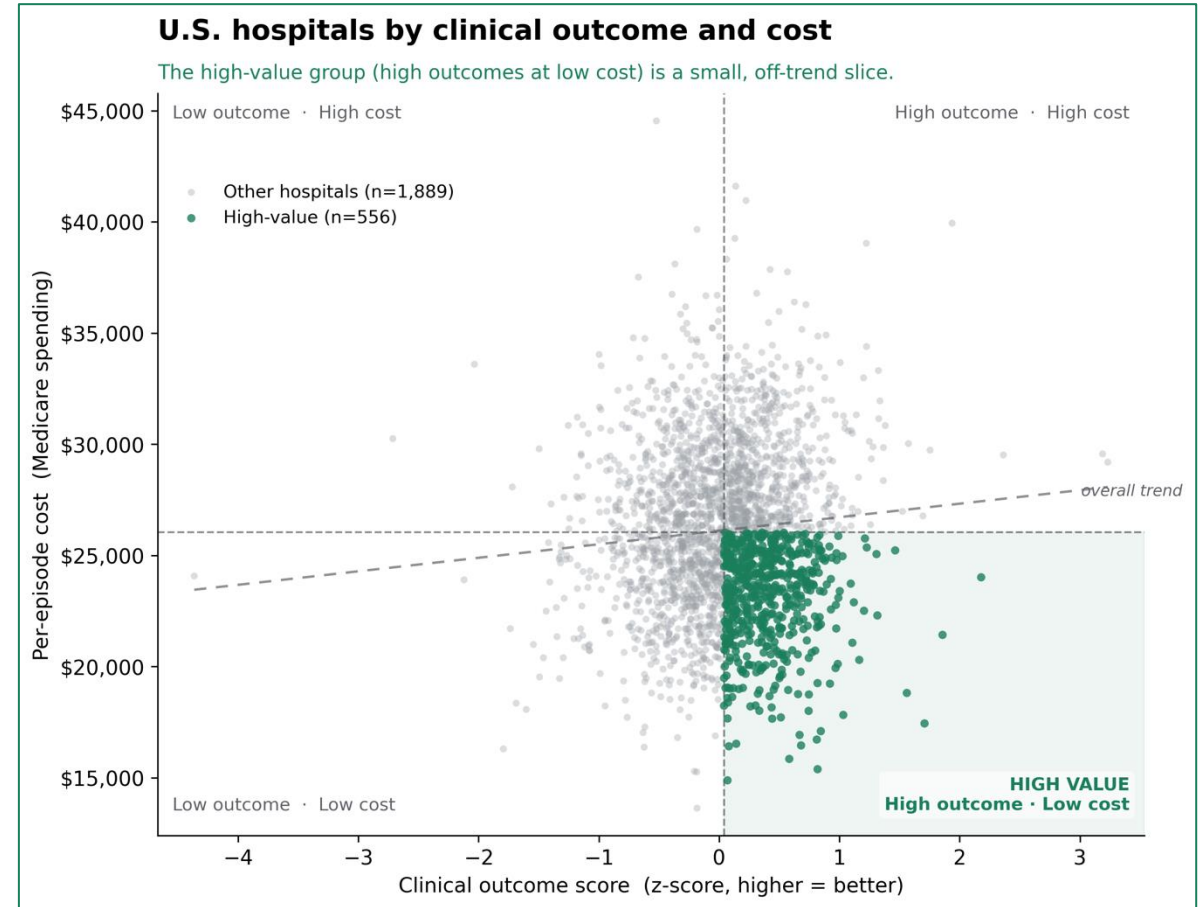
You don't get what you pay for

Higher cost buys only marginally better outcomes.

Because the two barely move together, the genuinely high-value hospitals (high outcomes at lower cost) are off-trend exceptions rather than the rule. They sit in the lower-right corner.

~528

high-value hospitals with above-median outcomes at below-median cost



Each dot is a hospital, plotted by clinical outcome and cost.

Four findings

1

Cost is predictable; clinical quality is not

Operations, patient experience, size, and location explain hospital cost moderately but barely explain outcomes. Happier patients don't necessarily have better clinical outcomes. Two independent statistical methods agree, so it's a limit of the data, not the model.

2

A cost screen works where a value screen can't

The tuned model identifies likely low-cost hospitals with ~84% precision on held-out data. This provides a usable operational filter, even though high value itself resists prediction.

3

The best hospitals fight the grain

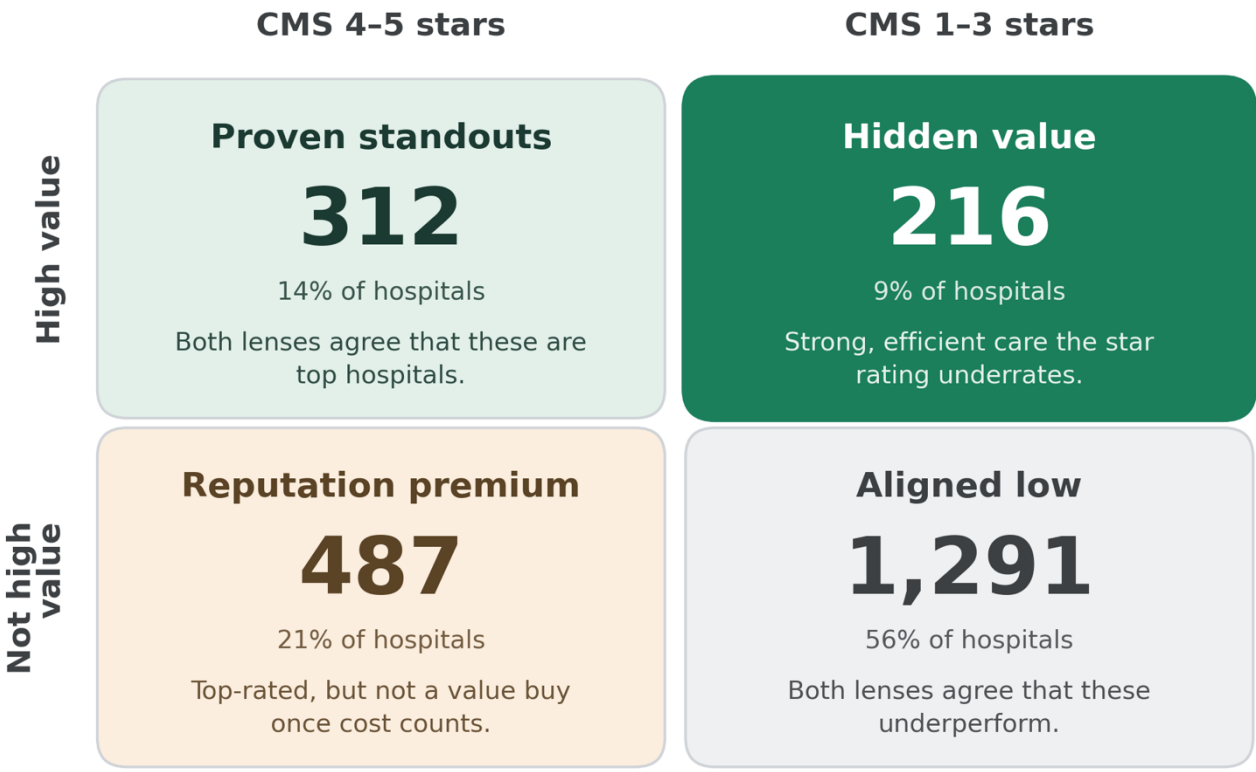
Outcomes and cost are mildly positively linked, so high-outcome / low-cost hospitals are genuine **efficiency outliers**. You can't spot exceptions from coarse proxies, so predictions fall short.

4

Stars and value agree ~70% of the time, but stars alone can't flag **value**

The 30% disagreement is concentrated exactly on the cost dimension the star ratings ignore.

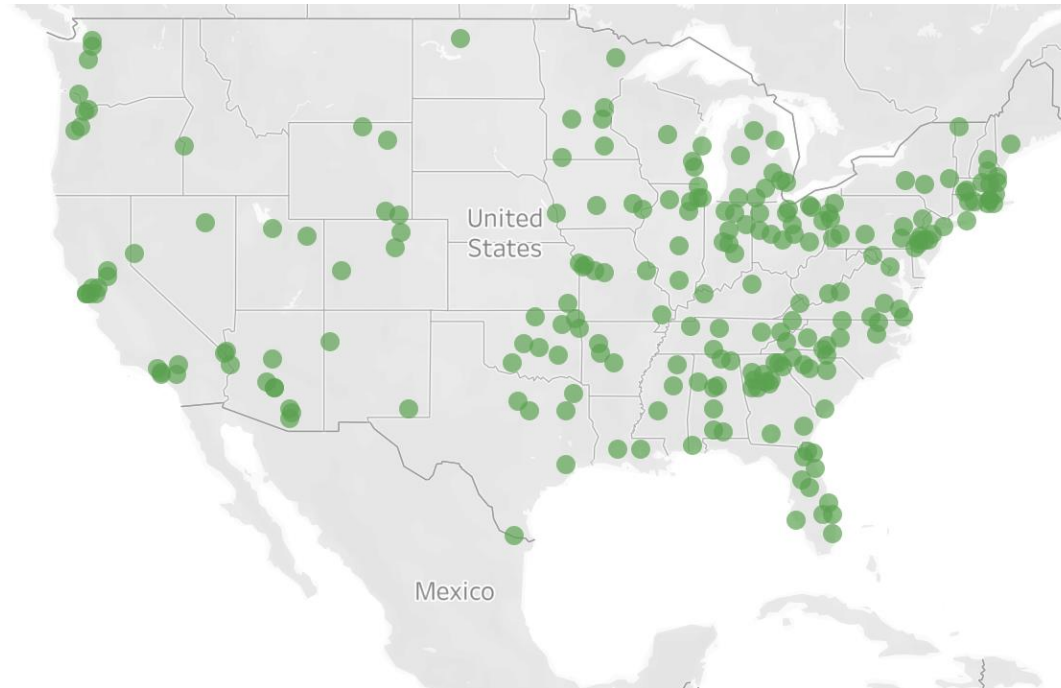
Stars can't see cost, so they miss some of the value



216
high-value hospitals rated only 1–3 stars, which translates to 2 in 5 of all high-value hospitals

61%
of top-rated (4–5 star) hospitals aren't high value once cost is counted

Where the high-value hidden hospitals are



Value tilts modestly by region and ownership

New England and non-profit systems lean favorable; some Southern divisions and government-run hospitals lean the other way. But variation within every group is large. The map shows where to look for underrated value, but it cannot give a **guarantee**.

Value must be **measured**, but cost can be **screened**

For payers, health systems, and care-coordination teams managing total cost of care, true high-value providers can't be inferred from price, reviews, or reputation. None of these reliably mirrors clinical quality. But the cost half can be screened quickly and reliably, and the hidden-value hospitals are exactly the **efficient** providers a star-only screen would miss.

Measure outcomes directly

Don't let low cost or strong reviews stand in as a proxy for clinical quality.

Screen cost cheaply

Deploy the low-cost model as an operational filter. With ~84% precision, it's ready to use.

Look past the stars

Surface the hidden-value providers the official ratings underrate.

The numbers behind the assertions

Two independent method families and two statistical targets both reach one consistent verdict: cost is moderately learnable, but clinical outcomes aren't. When models with very different assumptions agree, the limit is in the data, rather than the model.

	Outcomes	Cost
Linear regression (adj. R ²)	0.13	0.38
Random forest (ROC-AUC)	0.65	0.77

~84% precision

low-cost model, on a locked test set

~70% agreement

with official CMS star ratings (n = 2,306)

T A K E A W A Y

You may or may not get better outcomes with higher costs. And some of the best-value hospitals are overlooked by star ratings alone.



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